

Neelabh Raj Awasthi

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SUMMARY

Computer Science undergraduate with applied experience across machine learning, deep learning, and backend systems — spanning self-supervised representation learning, multimodal (vision + language) modeling, and production AI pipelines built with RAG, vector databases, and transformer models. Comfortable working across the stack, from model design and training to deployment behind REST APIs, with an emphasis on grounded, auditable, and scalable outputs. AWS Certified Solutions Architect – Associate; IBM-certified in Data Science.

TECHNICAL SKILLS

Machine Learning / DL: PyTorch, Deep Learning, Self-Supervised Learning (JEPA), Transformers, Representation Learning, Model Optimization, Scikit-learn
Vision & Language: Multimodal Modeling, Computer Vision, Embeddings, RAG, Vector Databases (ChromaDB), Prompt Engineering, NLP
Languages: Python, C, SQL, Bash · C++ / Java (coursework, familiar)
Data Science: Statistical Modeling, Feature Engineering, EDA, ML Pipeline Design, Data Visualization (Matplotlib, Seaborn)
Backend / Systems: Flask, REST APIs, Streamlit · Linux (Ubuntu, Debian), Git, Docker (familiar), Cloudflare Tunnels
Cloud: AWS (Solutions Architect – Associate), Cloudflare Tunnels / Zero Trust

WORK EXPERIENCE

ALBA IT Department, Bahrain **May 2025 – Jun 2025**
AI Engineering Intern

- Architected Fine Lines, an enterprise RAG document-intelligence system (ChromaDB) delivering source-cited, grounded policy answers at sub-second latency — a content-understanding pipeline for automated policy validation.
- Enforced multi-tenant safety and governance via department-scoped vector stores and role-based access control, ensuring grounded, auditable, and policy-compliant retrieval.
- Cut response latency by integrating the Groq API; added contextual follow-ups and TTS for real-time production performance.

Kraya and Kuber, KIIT Student Society **Jan 2025 – Apr 2025**
Marketing & Data Analytics Intern

- Drove data-driven outreach; tracked engagement metrics to iteratively optimize content strategy.

PROJECTS

Multimodal Content-Moderation Classifier **Jun 2026 – Aug 2026 (in progress)**

- Building a classifier fusing CLIP vision + text embeddings to flag policy-violating content; current benchmark: 0.63 F1 / 66% recall on the violating class over a 2,800-sample set.
- Evaluating multimodal vs. text-only signal fusion (+5 pt recall gain observed) to guide final model design.
- Serving via a FastAPI endpoint returning decision, predicted label, and confidence with per-request logging.

Latent World Model: Self-Supervised Future-State Prediction (JEPA) **Feb 2026 – Aug 2026 (in progress)**

- Designing and implementing a Joint-Embedding Predictive Architecture (JEPA) from scratch — a self-supervised deep learning model that predicts future states in a non-generative latent space.
- Building the training and optimization loop with an EMA-updated target encoder, engineering against representation (dimensional) collapse to ensure stable learned embeddings.
- Integrating with Gymnasium 2D physics environments for latent-space planning; targeting a 65% inference-latency reduction versus generative baselines.

Dynamic Pricing System for Retail **Dec 2025 – Mar 2026**

- Designed ML and rule-based pricing models driven by real-time demand and inventory signals.
- Benchmarked approaches against real-world adoption criteria; produced a comparative analysis to guide production deployment.

HR Onboarding Assistant (Amazon Q + RAG) **Aug 2025 – Sep 2025**

- Automated HR access-ticket workflows and onboarding comms via Amazon Q integration.
- Deployed an internal RAG pipeline for policy query resolution, reducing manual lookup time.

Self-Hosted Cloud & Media Server **Dec 2022 – Apr 2023**

- Deployed bare-metal Ubuntu with Nextcloud + Jellyfin; bypassed CG-NAT via Cloudflare Tunnels with Zero Trust for a zero-open-port, high-availability posture.

EDUCATION

B.Tech, Computer Science & Engineering — KIIT University **CGPA: 7.70 / 10 (till Sem 6)**
Class XII — Laureate Public School **91.2% · School Topper**

RELEVANT COURSEWORK

Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Design & Analysis of Algorithms, Machine Learning, Artificial Intelligence, Probability & Statistics, Linear Algebra, Software Engineering

CERTIFICATIONS

AWS Certified Solutions Architect – Associate · IBM Data Science Professional Certificate (Coursera) · Git & GitHub Fundamentals (Udemy)